

FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness

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ABSTRACT

Transformers are slow and memory-hungry on long sequences because self-attention scales quadratically with sequence length. We propose **FlashAttention**, an exact attention algorithm that uses tiling to compute softmax without materializing the large intermediate $N \times N$ matrix to GPU memory.

1. IO-Aware Tiling

FlashAttention splits inputs into blocks, loads them from HBM to SRAM, and performs incremental softmax. This reduces memory traffic by 5-10x and achieves 2-4x wall-clock speedups.

2. Quadratic Bottleneck

Standard attention incurs $O(N^2)$ memory reads and writes. FlashAttention reduces this to $O(N)$ memory reads/writes, scaling transformers to 64k+ context windows.

